Genome AMR analysis: SAMN11638310

Run status: completed Finished: 2026-10-13T19:00:02.837777+00:00 Report generated: 2026-10-13T19:03:10+00:00

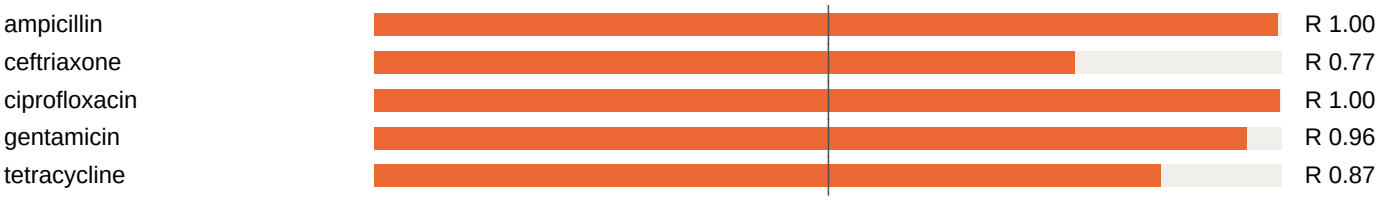
Research and educational use only. Not a clinical result; laboratory confirmation is required. Detected determinants and predicted phenotypes are separate outputs.

Input genome

Species	Contigs	Total length	N50	GC %	Ambiguous %
Escherichia coli (user-supplied)	115	5,549,710	120,573	50.62	0.000

File SHA-256: f5ea02bdb8080db9f6bd80c2d8b83d6ad0b11bb56c9b65f334428ce4aa361e7b

Predicted phenotypes (pretrained models)



Bars show resistance score; the line marks the 0.5 decision threshold.

Antibiotic	Prediction	Score	Status	Features	Note
ampicillin	R	0.9956	predicted	760/1000	-
ceftriaxone	R	0.7719	predicted	296/1000	-
ciprofloxacin	R	0.9975	predicted	363/1000	-
gentamicin	R	0.9616	predicted	565/1000	-
tetracycline	R	0.8666	predicted	28/1000	-
cefepime	-	-	unsupported	-	No validated model for this antibiotic.
trimethoprim/sulfamethoxazole	-	-	unsupported	-	No validated model for this antibiotic.

Scores are external model outputs, not locally calibrated probabilities.

Known resistance determinants (AMRFinderPlus)

37 known determinant elements detected.

Element	Type	Evidence	Class	Method	Identity %	Coverage %	Contig
gyrA_D87N	AMR	mutation	QUINOLONE	POINTX	98.74	99.66	DABTJL010000001.1
gyrA_S83L	AMR	mutation	QUINOLONE	POINTX	98.74	99.66	DABTJL010000001.1
ftsI_I336IKYRI	AMR	mutation	BETA-LACTAM	POINTX	99.83	100.0	DABTJL010000003.1
parC_S80I	AMR	mutation	QUINOLONE	POINTX	99.87	100.0	DABTJL010000004.1
parE_S458A	AMR	mutation	QUINOLONE	POINTX	99.68	100.0	DABTJL010000004.1
ompF_Y253EfsTer3	AMR	mutation	BETA-LACTAM	POINTX	92.56	100.0	DABTJL010000017.1
cyaA_S352T	AMR	mutation	FOSMIDOMYCIN	POINTX	99.29	100.0	DABTJL010000022.1
blaTEM-1	AMR	gene	BETA-LACTAM	ALLELEX	100.0	100.0	DABTJL010000073.1
blaNDM-5	AMR	gene	BETA-LACTAM	ALLELEX	100.0	100.0	DABTJL010000075.1
ble	AMR	gene	BLEOMYCIN	EXACTX	100.0	100.0	DABTJL010000075.1
blaCTX-M-15	AMR	gene	BETA-LACTAM	ALLELEX	100.0	100.0	DABTJL010000080.1
tet(B)	AMR	gene	TETRACYCLINE	EXACTX	100.0	100.0	DABTJL010000082.1
aadA2	AMR	gene	AMINOGLYCOSIDE	EXACTX	100.0	100.0	DABTJL010000084.1
dfrA12	AMR	gene	TRIMETHOPRIM	EXACTX	100.0	100.0	DABTJL010000084.1
catB3	AMR	gene	PHENICOL	PARTIAL_CONTIG_ENDX	100.0	69.52	DABTJL010000086.1
blaOXA-1	AMR	gene	BETA-LACTAM	ALLELEX	100.0	100.0	DABTJL010000086.1
aac(6')-Ib-cr5	AMR	gene	AMINOGLYCOSIDE/QUINOLONE	ALLELEX	100.0	100.0	DABTJL010000086.1
aac(3)-IId	AMR	gene	AMINOGLYCOSIDE	EXACTX	100.0	100.0	DABTJL010000088.1
aadA5	AMR	gene	AMINOGLYCOSIDE	EXACTX	100.0	100.0	DABTJL010000089.1

Element	Type	Evidence	Class	Method	Identity %	Coverage %	Contig
dfrA17	AMR	gene	TRIMETHOPRIM	EXACTX	100.0	100.0	DABTJL010000089.1
sul1	AMR	gene	SULFONAMIDE	EXACTX	100.0	100.0	DABTJL010000091.1
qacEdelta1	STRESS	other_element	QUATERNARY AMMONIUM	ALLELEX	100.0	100.0	DABTJL010000091.1
blaNDM-5	AMR	gene	BETA-LACTAM	ALLELEX	100.0	100.0	DABTJL010000107.1
ble	AMR	gene	BLEOMYCIN	EXACTX	100.0	100.0	DABTJL010000107.1
blaTEM-1	AMR	gene	BETA-LACTAM	ALLELEX	100.0	100.0	DABTJL010000108.1
blaCTX-M-15	AMR	gene	BETA-LACTAM	ALLELEX	100.0	100.0	DABTJL010000109.1
tet(B)	AMR	gene	TETRACYCLINE	EXACTX	100.0	100.0	DABTJL010000110.1
aac(6')-Ib-cr5	AMR	gene	AMINOGLYCOSIDE/QUINOLONE	ALLELEX	100.0	100.0	DABTJL010000111.1
blaOXA-1	AMR	gene	BETA-LACTAM	ALLELEX	100.0	100.0	DABTJL010000111.1
catB3	AMR	gene	PHENICOL	PARTIALX	100.0	70.0	DABTJL010000111.1
dfrA12	AMR	gene	TRIMETHOPRIM	EXACTX	100.0	100.0	DABTJL010000112.1
aadA2	AMR	gene	AMINOGLYCOSIDE	EXACTX	100.0	100.0	DABTJL010000112.1
dfrA17	AMR	gene	TRIMETHOPRIM	EXACTX	100.0	100.0	DABTJL010000113.1
aadA5	AMR	gene	AMINOGLYCOSIDE	EXACTX	100.0	100.0	DABTJL010000113.1
aac(3)-IId	AMR	gene	AMINOGLYCOSIDE	EXACTX	100.0	100.0	DABTJL010000114.1
qacEdelta1	STRESS	other_element	QUATERNARY AMMONIUM	ALLELEX	100.0	100.0	DABTJL010000115.1
sul1	AMR	gene	SULFONAMIDE	EXACTX	100.0	100.0	DABTJL010000115.1

Limitations

- Research and educational use only; laboratory confirmation is required.
- Known determinant detection and predicted phenotypes are separate outputs.
- No detected determinant does not establish susceptibility.
- E. coli species is user-supplied; automatic identification is unavailable.
- External models; not validated on our cohort.
- Training overlap with our NCBI/PATRIC cohort is unknown.
- Scores are not locally calibrated probabilities.
- Cefepime and trimethoprim/sulfamethoxazole unsupported.

Provenance

Package version: 0.1.0
Engine SHA-256: 63cd3fce93f5703ba770c519834371032a771cc884270db4e48b739a18adaac2
Configuration SHA-256: 1d8c79a54a5fe7585002b42f9c8661732246c98cd4730534226f2444807d5399
Detector version: 4.2.7
Detector database: 2026-08-07.1
Model registry SHA-256: 3cc237807c58733044a88b46f3995d43726703e45d0d23bc3129bfc2507fc2a2
analysis.json SHA-256: 8e5117f0392488628f3935bd931b2cc55f0450e8ddba56c83d03db6ca4c41f93